GAME DESIGN DOCUMENT

SUMMARY

- Title: The Turtle Labyrinth
- Developers: Celeste Bazzi, Christian Parodi, Dario Valentini

GAME PLAY

General Vison
"The Labf Turtle" is an educational augmented reality game that offers three different game modes: “Labyrinth”, which requires solving a maze; “Free mode”, where the user can freely draw on the terrain onto which Tilly the turtle moves; “Code mode”, where the movements of the turtle can be repeated through the use of while loops, helping the user understand this basic programming concept. Using an Android device, players can interact with Tilly the turtle in the real world to solve problems, create and learn.

Objective
The game provides an educational and entertaining experience through the use of augmented reality, combining creativity, computer knowledge and problem-solving skills.

Environmental Interaction
Use the Android device's camera to overlay game elements in the real world.

MECHANICS

General Mechanics

- *AR navigation and interaction*: the movements of Tilly are controlled by virtual buttons that appear on the display of the Android device.
- *LOGO language learning*: each action performed by Tilly is written on a floating board right next to the game map, following the semantics of the LOGO language. We implemented a simplified version of LOGO, which consists of just 3 commands, “Forward”, “Right 45” and “Left 45”.

Labyrinth Mode

Players guide Tilly through a maze to arrive at the finish, marked with a colorful circle.

Free Mode

Players can use the buttons to trace lines and shapes on the virtual map inside the real world. A new button is introduced, to make Tilly start/stop leaving colorful trails as she moves around.

Code Mode

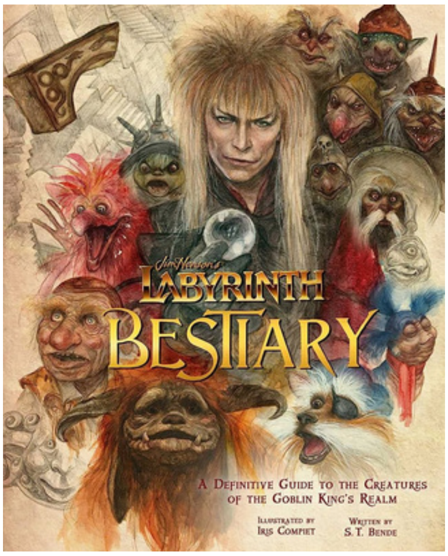
A simple while loop mechanics is implemented. The player can queue multiple LOGO commands and then run them for a chosen number of repetitions, as if they were in a while loop. The floating board will display the commands, with an appropriate layout useful to understand the while loop. After clicking on the “Run” button, Tilly the turtle will follow the movements defined by the selected commands, creating a drawing.

STRUCTURE

App structure
The initial hierarchy consists of the AR camera and the image target; when the target is detected, the game map spawns upon it and arrow buttons appear on the screen. In every mode, the game ends either when Tilly reaches the finish or the player taps the home button in the upper left corner.
A script attached to Tilly is responsible for her movements, which are triggered by the click of the buttons.

Components

- *Box Colliders*: used to avoid collision with walls;
- *Trail Renderer*: used to have the turtle leave a trail behind her in Free and Code modes.



Target image used

Main libraries

- *Vuforia*: AR engine used. The main usage is to identify the target image and generete the augmentation upon it;
- *AI Navigation*: useful to constraint the movements of the turtle to the areas allowed. This way, she can't cross walls, or go through neither bushes nor trees;
- *AR Foundation*
- *Google ARCore*
- *TextMeshPro*
- *XR Interaction Toolkit*

ASSETS

Graphics
We used several free assets to create the environment (tree, grass, terrain...), then combined them together into different prefabs to create the game map. The turtle has been shaped in Blender and the splash screens, like the Main Menu and the "You won!" screen, have been made in Canva.

User Interface & Elements

- *Character*: Tilly the turtle is the main character controlled by the player;
- *Objects*: labyrinth for Labyrinth mode, no-walls map for Free and Code mode, Scrollview to display LOGO language commands;
- *Controls*: buttons appear on a canvas that lies on the Screen space.
- *Main screen*: lets the player choose the game mode.

Tutorial
Below there is the link to view the video tutorial for the game of "The Turtle Labyrinth", together with the. APK file and the UnityPackage.

Google Drive Folder